

LLDD BE - API Report Master Config

SBP Mall - ระบบประกันรายได้ | Low Level Design Document

แนวทางการใช้เอกสาร

หัวข้อ	คำอธิบาย
วัตถุประสงค์	ใช้เป็นรายละเอียดระดับพัฒนาสำหรับออกแบบ ลงมือพัฒนา ตรวจสอบ และทดสอบขอบเขตที่ระบุในฉบับนี้
ลำดับการอ่าน	เริ่มจาก Overview และ Scope จากนั้นตรวจ Field/Validation, Implementation, Contract, Processing Flow, Acceptance Criteria และ Test Checklist ตามลำดับ
ผลลัพธ์ที่คาดหวัง	ผู้อ่านสามารถระบุ input, ขั้นตอนประมวลผล, output, เงื่อนไขผิดพลาด และหลักฐานการทดสอบได้จากเนื้อหาในฉบับเดียว

คำว่า Input, Progress และ Output ในเอกสารนี้หมายถึงข้อมูลตั้งต้น ลำดับการทำงาน และผลลัพธ์ที่ตรวจสอบได้ของขอบเขตที่กำลังพัฒนา

1. Overview

รายการ	รายละเอียด
Track	BE
Estimate	30 ชั่วโมง
Owner	Tunyatorn <Vava> Kiatkongphongsa
Objective	ออกแบบ APIs สำหรับรายงาน Master Data และ System Config

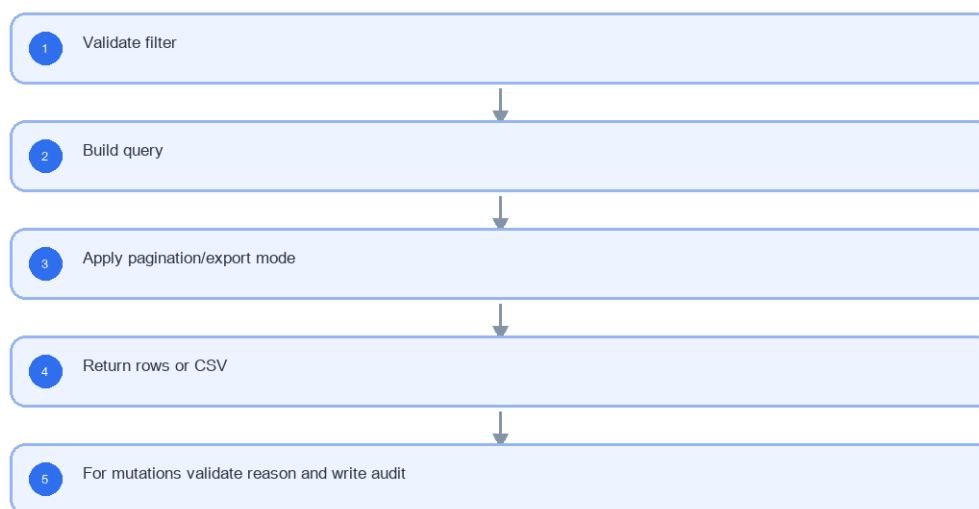
Common contract reference: ทุกหัวข้อ API/FE ต้องยึด LLDD-BE-API-Common-Contracts.pdf และ LLDD-FE-Integration-Contracts.pdf สำหรับ error/auth/format/pagination/action/RBAC ก่อนลงรายละเอียดเฉพาะหน้าหรือเฉพาะ endpoint

2. Screen / Functional Scope

- Report query service
- CSV export
- Operator/factor CRUD
- System/Global Config (SCR-11)
- Report filters

4. Implementation Flow Diagram (Reference)

LLDD BE - API Report Master Config



รูปที่ 1: Implementation flow reference: LLDD BE - API Report Master Config

5. Field, Format, and Validation

Field / UI	Format	Validation	Behavior
year	พ.ศ. YYYY	required for report	return 400 if missing
status	statusCode string	required	6 สถานะเอกสาร; verbatim จาก document_statuses
result	APPROVE REJECT	required for report	maps to consideration latest result

Field / UI	Format	Validation	Behavior
region	array/string	optional	13 region codes; multi-select
storeType	array/string	optional	A/B/C/E; multi-select
impactedStoreCode	string 5 digits	optional	คง leading zero
newStoreCode	string 5 digits	optional	คง leading zero
reason	text	required mutation	audit reason
page/size	integer	page>=1 size<=100	pagination

5.1 Input / Progress / Output Contract

Stage	Contract for implementation
Input	GET /api/v1/reports/status-summary; GET /api/v1/reports/status-summary/export; GET /api/v1/operators
Progress	Validate filter; Build query; Apply pagination/export mode; Return rows or CSV
Output	operator_assignments; external_factors; mas_param (SBP)

5.90 Endpoint Implementation Contract

Endpoint	Use-case owner	Service/repository behavior	Definition of done
GET /api/v1/reports/status-summary	รายงานตรวจสอบประกันรายได้	Validate filter	missing year/status/result fails
GET /api/v1/reports/status-summary/export	Export Excel	Build query	export uses same filters as preview
GET /api/v1/operators	อ่าน operator assignments	Apply pagination/export mode	master edit requires reason
POST /api/v1/operators	สร้าง operator assignment	Return rows or CSV	config locked value cannot edit
PUT /api/v1/operators/{id}	แก้ operator assignment	For mutations validate reason and write audit	missing year/status/result fails
DELETE /api/v1/operators/{id}	ยกเลิก operator assignment	Validate filter	export uses same filters as preview
GET /api/v1/factors	อ่านปัจจัยภายนอก	Build query	master edit requires reason
POST /api/v1/factors	สร้างปัจจัยภายนอก	Apply pagination/export mode	config locked value cannot edit
PUT /api/v1/factors/{code}	แก้ปัจจัยภายนอก	Return rows or CSV	missing year/status/result fails
DELETE /api/v1/factors/{code}	ลบปัจจัยภายนอกที่ไม่ถูกใช้งาน	For mutations validate reason and write audit	export uses same filters as preview
GET /api/v1/configs	อ่าน system config	Validate filter	master edit requires reason
GET /api/v1/configs/{key}	อ่าน config ราย key	Build query	config locked value cannot edit
POST /api/v1/configs	สร้าง config ที่ไม่ใช่ secret	Apply pagination/export mode	missing year/status/result fails
PUT /api/v1/configs/{key}	แก้ config ที่ editable=true	Return rows or CSV	export uses same filters as preview
DELETE /api/v1/configs/{key}	ลบ config ที่ editable=true และไม่ถูกใช้งาน	For mutations validate reason and write audit	master edit requires reason

5.91 Backend Execution Sequence

Step	Behavior specific to this LLDD	Failure/test evidence
1	Validate filter	report missing year
2	Build query	report export

Step	Behavior specific to this LLDD	Failure/test evidence
3	Apply pagination/export mode	factor duplicate
4	Return rows or CSV	operator audit
5	For mutations validate reason and write audit	config locked

6. Button / User Action Mapping

Action	Trigger	API / Service	Expected Result
Report preview	GET	report.service.search	paginated rows
Report export	GET	report.service.exportCsv	csv stream
Master mutation	POST/PUT/DELETE	master.service.save	audit log

7. API Contract

GET /api/v1/reports/status-summary

รายงานตรวจสอบประกันรายได้

Query Params
<pre>{ "year": 2026, "status": "06", "result": "APPROVE", "region": ["RSU"], "storeType": ["A"], "impactedStoreCode": "00788", "newStoreCode": "00990", "page": 1, "size": 20 }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
year	integer	Yes	UTF-8; use value domain described by endpoint purpose
status	string	Yes	UTF-8; use value domain described by endpoint purpose
result	string	No	UTF-8; use value domain described by endpoint purpose
region	array<string>	No	JSON array; element type shown in Type column
storeType	array<string>	No	JSON array; element type shown in Type column
impactedStoreCode	string	No	exactly 5 digits; preserve leading zero
newStoreCode	string	No	exactly 5 digits; preserve leading zero
page	integer	No	>= 1; default 1
size	integer	No	1..100; default 20

Response
<pre>{ "page": 1, "size": 20, "total": 0, "items": [] }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
page	integer	Yes	>= 1; default 1
size	integer	Yes	1..100; default 20
total	integer	Yes	UTF-8; use value domain described by endpoint purpose
items	array<object>	Yes	JSON array; element type shown in Type column

GET /api/v1/reports/status-summary/export

Export Excel

Query Params
<pre>{ "year": 2026, "status": "06", "result": "APPROVE", "region": ["RSU"], "storeType": ["A"], "impactedStoreCode": "00788", "newStoreCode": "00990" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
year	integer	Yes	UTF-8; use value domain described by endpoint purpose
status	string	Yes	UTF-8; use value domain described by endpoint purpose
result	string	No	UTF-8; use value domain described by endpoint purpose
region	array<string>	No	JSON array; element type shown in Type column
storeType	array<string>	No	JSON array; element type shown in Type column
impactedStoreCode	string	No	exactly 5 digits; preserve leading zero
newStoreCode	string	No	exactly 5 digits; preserve leading zero

Response

```
{
  "fileName": "insurance-verification-2026.xlsx"
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
fileName	string	Yes	UTF-8; use value domain described by endpoint purpose

GET /api/v1/operators

อ่าน operator assignments

Query Params

```
{
  "employeeId": "E001",
  "positionCode": "06",
  "active": true,
  "page": 1,
  "size": 20
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
employeeId	string	No	UTF-8; use value domain described by endpoint purpose
positionCode	string	No	UTF-8; use value domain described by endpoint purpose
active	boolean	No	UTF-8; use value domain described by endpoint purpose
page	integer	No	>= 1; default 1
size	integer	No	1..100; default 20

Response

```
{
  "page": 1,
  "size": 20,
  "total": 1,
  "items": [
    {
      "id": 101,
      "employeeId": "E001",
      "employeeName": "สมชาย ใจดี",
      "positionCode": "06",
      "zoneCode": "01",
      "active": true
    }
  ]
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
page	integer	Yes	>= 1; default 1
size	integer	Yes	1..100; default 20
total	integer	Yes	UTF-8; use value domain described by endpoint purpose

Field	Type	Required	Constraint / Meaning
items	array<object>	Yes	JSON array; element type shown in Type column
items[].id	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].employeeId	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].employeeName	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].positionCode	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].zoneCode	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].active	boolean	Yes	UTF-8; use value domain described by endpoint purpose

POST /api/v1/operators

สร้าง operator assignment

Request
<pre>{ "employeeId": "E001", "positionCode": "06", "zoneCode": "01", "active": true, "reason": "มอบหมายผู้ปฏิบัติงาน" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
employeeId	string	Yes	UTF-8; use value domain described by endpoint purpose
positionCode	string	Yes	UTF-8; use value domain described by endpoint purpose
zoneCode	string	Yes	UTF-8; use value domain described by endpoint purpose
active	boolean	Yes	UTF-8; use value domain described by endpoint purpose
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response
<pre>{ "id": 101, "employeeId": "E001", "positionCode": "06", "zoneCode": "01", "active": true }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
id	integer	Yes	UTF-8; use value domain described by endpoint purpose

Field	Type	Required	Constraint / Meaning
employeeId	string	Yes	UTF-8; use value domain described by endpoint purpose
positionCode	string	Yes	UTF-8; use value domain described by endpoint purpose
zoneCode	string	Yes	UTF-8; use value domain described by endpoint purpose
active	boolean	Yes	UTF-8; use value domain described by endpoint purpose

PUT /api/v1/operators/{id}

แก้ operator assignment

Request
<pre>{ "positionCode": "08", "zoneCode": "01", "active": true, "reason": "ย้ายหน้าที่" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
positionCode	string	Yes	UTF-8; use value domain described by endpoint purpose
zoneCode	string	Yes	UTF-8; use value domain described by endpoint purpose
active	boolean	Yes	UTF-8; use value domain described by endpoint purpose
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response
<pre>{ "id": 101, "employeeId": "E001", "positionCode": "08", "zoneCode": "01", "active": true }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
id	integer	Yes	UTF-8; use value domain described by endpoint purpose
employeeId	string	Yes	UTF-8; use value domain described by endpoint purpose
positionCode	string	Yes	UTF-8; use value domain described by endpoint purpose
zoneCode	string	Yes	UTF-8; use value domain described by endpoint purpose
active	boolean	Yes	UTF-8; use value domain described by endpoint purpose

DELETE /api/v1/operators/{id}

ยกเลิก operator assignment

Request

```
{
  "reason": "สิ้นสุดการมอบหมาย"
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response

```
{
  "id": 101,
  "deleted": true
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
id	integer	Yes	UTF-8; use value domain described by endpoint purpose
deleted	boolean	Yes	UTF-8; use value domain described by endpoint purpose

GET /api/v1/factors

อ่านปัจจัยภายนอก

Query Params

```
{
  "q": "ก่อสร้าง",
  "active": true,
  "page": 1,
  "size": 20
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
q	string	No	UTF-8; use value domain described by endpoint purpose
active	boolean	No	UTF-8; use value domain described by endpoint purpose
page	integer	No	>= 1; default 1
size	integer	No	1..100; default 20

Response

```
{
  "page": 1,
  "size": 20,
  "total": 1,
  "items": [
    {
      "factorCode": "ROAD",
      "factorName": "ก่อสร้างถนน",
      "description": "ปิดช่องทางจราจร",
      "active": true
    }
  ]
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
page	integer	Yes	>= 1; default 1
size	integer	Yes	1..100; default 20
total	integer	Yes	UTF-8; use value domain described by endpoint purpose
items	array<object>	Yes	JSON array; element type shown in Type column
items[].factorCode	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].factorName	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].description	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].active	boolean	Yes	UTF-8; use value domain described by endpoint purpose

POST /api/v1/factors

สร้างปัจจัยภายนอก

Request

```
{
  "factorCode": "ROAD",
  "factorName": "ก่อสร้างถนน",
  "description": "ปิดช่องทางจราจร",
  "active": true,
  "reason": "เพิ่มปัจจัยใหม่"
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
factorCode	string	Yes	UTF-8; use value domain described by endpoint purpose
factorName	string	Yes	UTF-8; use value domain described by endpoint purpose
description	string	Yes	UTF-8; use value domain described by endpoint purpose
active	boolean	Yes	UTF-8; use value domain described by endpoint purpose

Field	Type	Required	Constraint / Meaning
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response

```
{
  "factorCode": "ROAD",
  "factorName": "ก่อสร้างถนน",
  "active": true
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
factorCode	string	Yes	UTF-8; use value domain described by endpoint purpose
factorName	string	Yes	UTF-8; use value domain described by endpoint purpose
active	boolean	Yes	UTF-8; use value domain described by endpoint purpose

PUT /api/v1/factors/{code}

แก้ไขปัจจัยภายนอก

Request

```
{
  "factorName": "ก่อสร้างและปิดถนน",
  "description": "มีดช่องทางจราจรบางส่วน",
  "active": true,
  "reason": "ปรับค่าอธิบาย"
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
factorName	string	Yes	UTF-8; use value domain described by endpoint purpose
description	string	Yes	UTF-8; use value domain described by endpoint purpose
active	boolean	Yes	UTF-8; use value domain described by endpoint purpose
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response

```
{
  "factorCode": "ROAD",
  "factorName": "ก่อสร้างและปิดถนน",
  "active": true
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
factorCode	string	Yes	UTF-8; use value domain described by endpoint purpose
factorName	string	Yes	UTF-8; use value domain described by endpoint purpose
active	boolean	Yes	UTF-8; use value domain described by endpoint purpose

DELETE /api/v1/factors/{code}

ลบปัจจัยภายนอกที่ไม่ถูกใช้งาน

Request
<pre>{ "reason": "ยกเลิกค่าทดสอบ" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response
<pre>{ "factorCode": "ROAD", "deleted": true }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
factorCode	string	Yes	UTF-8; use value domain described by endpoint purpose
deleted	boolean	Yes	UTF-8; use value domain described by endpoint purpose

GET /api/v1/configs

อ่าน system config

Query Params
<pre>{ "q": "UPLOAD", "page": 1, "size": 20 }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
q	string	No	UTF-8; use value domain described by endpoint purpose
page	integer	No	>= 1; default 1

Field	Type	Required	Constraint / Meaning
size	integer	No	1..100; default 20

Response

```
{
  "page": 1,
  "size": 20,
  "total": 1,
  "items": [
    {
      "key": "UPLOAD_MAX_MB",
      "value": 5,
      "valueType": "INTEGER",
      "editable": true,
      "description": "ขนาดไฟล์สูงสุด"
    }
  ]
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
page	integer	Yes	>= 1; default 1
size	integer	Yes	1..100; default 20
total	integer	Yes	UTF-8; use value domain described by endpoint purpose
items	array<object>	Yes	JSON array; element type shown in Type column
items[].key	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].value	integer	Yes	UTF-8; use value domain described by endpoint purpose
items[].valueType	string	Yes	UTF-8; use value domain described by endpoint purpose
items[].editable	boolean	Yes	UTF-8; use value domain described by endpoint purpose
items[].description	string	Yes	UTF-8; use value domain described by endpoint purpose

GET /api/v1/configs/{key}

อ่าน config ราย key

Query Params

```
{
  "key": "UPLOAD_MAX_MB"
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
key	string	No	UTF-8; use value domain described by endpoint purpose

Response
<pre>{ "key": "UPLOAD_MAX_MB", "value": 5, "valueType": "INTEGER", "editable": true, "description": "ขนาดไฟล์สูงสุด" }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
key	string	Yes	UTF-8; use value domain described by endpoint purpose
value	integer	Yes	UTF-8; use value domain described by endpoint purpose
valueType	string	Yes	UTF-8; use value domain described by endpoint purpose
editable	boolean	Yes	UTF-8; use value domain described by endpoint purpose
description	string	Yes	UTF-8; use value domain described by endpoint purpose

POST /api/v1/configs

สร้าง config ที่ไม่ใช่ secret

Request
<pre>{ "key": "UPLOAD_WARN_MB", "value": 4, "valueType": "INTEGER", "editable": true, "description": "ระดับเตือนขนาดไฟล์", "reason": "เพิ่มค่าเตือน" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
key	string	Yes	UTF-8; use value domain described by endpoint purpose
value	integer	Yes	UTF-8; use value domain described by endpoint purpose
valueType	string	Yes	UTF-8; use value domain described by endpoint purpose
editable	boolean	Yes	UTF-8; use value domain described by endpoint purpose
description	string	Yes	UTF-8; use value domain described by endpoint purpose
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response

```
{
  "key": "UPLOAD_WARN_MB",
  "value": 4,
  "valueType": "INTEGER",
  "editable": true
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
key	string	Yes	UTF-8; use value domain described by endpoint purpose
value	integer	Yes	UTF-8; use value domain described by endpoint purpose
valueType	string	Yes	UTF-8; use value domain described by endpoint purpose
editable	boolean	Yes	UTF-8; use value domain described by endpoint purpose

PUT /api/v1/configs/{key}

แก้ config ที่ editable=true

Request

```
{
  "value": 6,
  "description": "ขนาดไฟล์สูงสุด",
  "reason": "ปรับตามนโยบายไฟล์"
}
```

Request Field Schema

Field	Type	Required	Constraint / Meaning
value	integer	Yes	UTF-8; use value domain described by endpoint purpose
description	string	Yes	UTF-8; use value domain described by endpoint purpose
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response

```
{
  "key": "UPLOAD_MAX_MB",
  "value": 6,
  "valueType": "INTEGER",
  "editable": true
}
```

Response Field Schema

Field	Type	Required	Constraint / Meaning
key	string	Yes	UTF-8; use value domain described by endpoint purpose
value	integer	Yes	UTF-8; use value domain described by endpoint purpose

Field	Type	Required	Constraint / Meaning
valueType	string	Yes	UTF-8; use value domain described by endpoint purpose
editable	boolean	Yes	UTF-8; use value domain described by endpoint purpose

DELETE /api/v1/configs/{key}

ลบ config ที่ editable=true และไม่ถูกใช้งาน

Request
<pre>{ "reason": "ยกเลิกค่าทดสอบ" }</pre>

Request Field Schema

Field	Type	Required	Constraint / Meaning
reason	string	Yes	trimmed UTF-8 Thai text; required by operation/business rule

Response
<pre>{ "key": "UPLOAD_WARN_MB", "deleted": true }</pre>

Response Field Schema

Field	Type	Required	Constraint / Meaning
key	string	Yes	UTF-8; use value domain described by endpoint purpose
deleted	boolean	Yes	UTF-8; use value domain described by endpoint purpose

8. Reference DB Mapping (No Database Page Work)

ส่วนนี้เป็นข้อมูลอ้างอิงสำหรับการ implement API/Job เท่านั้น ไม่ใช้งานสร้างหน้า Database, ไม่ใช้งานออกแบบ DB page และไม่ถูกนับเป็น deliverable แยกของ FE/BE

Table / Object	R/W	Usage
compensation_documents	R	แหล่งข้อมูลรายงานและ filter status/year
compensation_histories	R	ยอดเงินชดเชยและงวด statement
consideration_logs	R	ผลพิจารณาล่าสุด APPROVE/REJECT
operator_assignments	R/W	ผู้ปฏิบัติงาน
external_factors	R/W	master บัญชีภายนอก
mas_param (SBP)	R/W	ค่ากำหนดกลางในตารางของระบบ SBP เดิม
audit_logs	W	ประวัติการแก้ไข operator/factor/config

9. Processing Flow

Step	Description
1	Validate filter
2	Build query
3	Apply pagination/export mode
4	Return rows or CSV
5	For mutations validate reason and write audit

10. Acceptance Criteria

- missing year/status/result fails
- export uses same filters as preview
- master edit requires reason
- config locked value cannot edit

11. Developer Test Checklist

No	Test
1	report missing year
2	report export
3	factor duplicate
4	operator audit
5	config locked